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RESEARCH INTERESTS

1. Post-training Alignment for Trustworthy LLMs

Safety Alignment, Preference Learning

2. Robust and Reliable LLM Inference

Hallucination Mitigation, Jailbreak Detection, OOD Detection, Multimodal Robustness, Model Selection & Routing

3. Trustworthy LLM Agents and Agentic Systems

Agent Safety, Runtime Reliability, Multi-Agent Orchestration

EDUCATION

University of Southern California, Los Angeles, CA

Aug 2024 – Present

Ph.D. in Computer Science

· Affiliation: **FORTIS** Lab; Advisor: Prof. Yue Zhao

University of Southern California, Los Angeles, CA

Jan 2022 – Dec 2023

M.S. in Machine Learning and Data Science

Nankai University, Tianjin, China

Sep 2017 – Jul 2021

B.E. in Software Engineering

WORK EXPERIENCE

LinkedIn, AI/ML Engineer Intern, Generative AI

May 2026 – Aug 2026

User Simulation Agent, LLM Reasoning, Post-training

Sunnyvale, CA

- Developing the reasoning component of an LLM-based user-simulation agent to improve user-behavior prediction, via post-training with GRPO and OPSD.

SELECTED PREPRINTS & UNDER SUBMISSION

* denotes joint first authors. See *Google Scholar* for the full list.

· **Cat-DPO: Category-Adaptive Safety Alignment**

Tiankai Yang*, Yi Nian*, Xinyuan Li, Ruiyao Xu, Henry Peng Zou, Kaize Ding, Xiyang Hu, Yan Liu, Yue Zhao.
Under submission. arXiv preprint arXiv:2604.17299

· **No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents**

Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan A. Rossi, Kaize Ding, Yue Zhao.
Under submission. arXiv preprint arXiv:2604.01350

· **HyperSkill: Self-Evolving LLM Agents via Hypergraph-Structured Skill Memory**

Ruiyao Xu, Tiankai Yang, Wei-Chieh Huang.
Under submission. arXiv preprint arXiv:2608.16114

SELECTED PUBLICATIONS

* denotes joint first authors. See [Google Scholar](#) for the full publication list.

- **CoAct: Co-Active LLM Preference Learning with Human-AI Synergy**

Ruiyao Xu, Mihir Parmar, [Tiankai Yang](#), Zhengyu Hu, Yue Zhao, Kaize Ding.
ACL, 🏆 Oral, 2026 📄

- **DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment**

Yi Nian, [Tiankai Yang](#), Yudi Zhang, Qi Pan, Zelong Xu, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, Yue Zhao.
EMNLP Findings, 2026

- **FlexRouter: Learning Complementary Model Sets for Flexible LLM Routing**

Wang Wei, Hongzheng Yang, [Tiankai Yang](#), Samyadeep Basu, Hongjie Chen, Andy Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry.
COLM, 2026

- **StealthRank: LLM Ranking Manipulation via Stealthy Prompt Optimization**

Yiming Tang, Yi Fan, Chenxiao Yu, [Tiankai Yang](#), Yue Zhao, Xiyang Hu.
Trustworthy AI for Good (AI4GOOD) Workshop @ ICML, 2026 📄

- **AD-AGENT: A Multi-agent Framework for End-to-end Anomaly Detection**

[Tiankai Yang*](#), Junjun Liu*, Michael Siu*, Jiahang Wang, Zhuangzhuang Qian, Chanjuan Song, Cheng Cheng, Xiyang Hu, Yue Zhao.
IJCNLP-AAACL Findings, 2025 📄

- **Treble Counterfactual VLMs: A Causal Approach to Hallucination**

Shawn Li, Jiashu Qu, Yuxiao Zhou, Yuehan Qin, [Tiankai Yang](#), Yue Zhao.
EMNLP Findings, 2025 📄

- **AD-LLM: Benchmarking Large Language Models for Anomaly Detection**

[Tiankai Yang*](#), Yi Nian*, Shawn Li, Ruiyao Xu, Yuangang Li, Jiaqi Li, Zhuo Xiao, Xiyang Hu, Ryan A. Rossi, Kaize Ding, Xia Hu, Yue Zhao.
ACL Findings, 2025 📄

- **DPU: Dynamic Prototype Updating for Multimodal Out-of-Distribution Detection**

Shawn Li, Huixian Gong, Hao Dong, [Tiankai Yang](#), Zhengzhong Tu, Yue Zhao.
CVPR, 🏆 Highlight, 2025 📄

- **PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection**

Sihan Chen*, Zhuangzhuang Qian*, Michael Siu*, [Tiankai Yang](#), Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, Yue Zhao.
The Web Conference (Demo Track), 2025 📄

SERVICES

Program Committee for Conferences

- NeurIPS 2026; ACL ARR 2025 (*Outstanding Reviewer*), 2026; AAAI 2026, 2027; COLM 2026; ICLR 2026; ICML 2025, 2026.

Journal Reviewer

- IEEE Transactions on Computational Social Systems